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An Explainable Stacking Framework for Multiclass Recorded Sleep-Status Classification

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Abstract

Small tabular health benchmarks can yield optimistic estimates when duplicate predictor profiles cross validation boundaries. We re-evaluated Random Forest–XGBoost stacking on the public Sleep Health and Lifestyle Dataset (374 records) after auditing outcome coding and duplicate structure. The source file contains 219 literal "None", 77 "Insomnia", and 78 "Sleep Apnea" labels, with no blank outcome values. After identifier removal and deterministic preparation, 265 rows duplicated a retained predictor profile and 22 of 109 profiles had conflicting recorded outcomes. We therefore used two repetitions of five-fold stratified group validation and generated stacking meta-features by four-fold group-aware out-of-fold fitting with preprocessing refit in each fold. No test-fold tuning was performed. Mean outer-fold accuracy (standard deviation) was 0.887 (0.034) for stacking, 0.890 (0.033) for soft voting, and 0.896 (0.032) for logistic regression; macro-F1 values were 0.832, 0.835, and 0.852, respectively. The mean stacking-minus-soft-voting accuracy difference was -0.0031 (bootstrap 95% interval -0.0162 to 0.0070), and the first-repeat McNemar test was non-significant ($p = 0.6875$). The contribution is a reproducible benchmark audit for recorded dataset labels; it is not clinical screening or diagnostic evidence.

1. Introduction

Sleep disorders represent a heterogeneous group of conditions that differ in clinical definition, underlying mechanisms, and management requirements. The International Classification of Sleep Disorders distinguishes insomnia disorders from sleep-related breathing disorders, including obstructive sleep apnea, and therefore a computational model that groups such conditions into a small number of benchmark categories should not be interpreted as reproducing the clinical taxonomy [1]. Nevertheless, automated risk-stratification approaches are increasingly investigated as tools for identifying individuals who may warrant further assessment. In particular, artificial-intelligence methods have been explored for sleep-apnea screening using electrocardiography, oximetry, respiratory signals, wearable sensors, and multimodal physiological measurements [3, 4, 11, 12]. These studies demonstrate the potential of machine learning to support scalable screening, while also highlighting the distinction between computational discrimination and clinically established diagnosis.

Alongside signal-based approaches, structured tabular sleep-health datasets provide a simpler environment for studying machine-learning methodology. Demographic characteristics, sleep duration and quality, physical activity, perceived stress, body-mass-index category, heart rate, blood pressure, and occupation can be combined to construct predictive models without requiring continuous physiological waveforms. Such datasets are attractive because they are inexpensive to analyse and readily accessible for methodological benchmarking. However, their simplicity can conceal important limitations. Outcome categories may not correspond to clinician-adjudicated diagnoses, predictor variables can be strongly correlated, class frequencies may be unequal, and administrative or preprocessing operations can unintentionally introduce information leakage. Consequently, apparently strong performance on a small tabular benchmark should be interpreted primarily as evidence of predictive behaviour within that dataset rather than as evidence of clinical validity.

Recent work has applied both conventional machine-learning ensembles and neural architectures to structured prediction problems. For tabular data, comparative evidence indicates that tree-based ensembles and deep neural networks can exhibit different strengths depending on the dataset characteristics and learning task [13]. Of particular relevance to the present study, Alom et al. evaluated ensemble neural-network methods on the same Sleep Health and Lifestyle Dataset considered here and reported accuracy values for bagging, boosting, and weighted-averaging configurations [14]. These findings provide a direct literature context for the present benchmark, although the two studies do not necessarily share identical preprocessing, data partitioning, validation, or model-selection procedures. The reported numerical results are

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therefore used as descriptive reference points rather than as evidence from a matched experimental comparison.

A further methodological concern arises when several predictive models are combined through stacking. Stacked generalization can exploit complementary errors and probability estimates from heterogeneous base learners, but its validity depends critically on how the second-level training data are constructed [28]. If a meta-learner is trained using in-sample predictions from base learners, information from observations used to fit those learners can propagate into the second stage and produce optimistic performance estimates. Similar concerns apply to preprocessing: scaling, categorical encoding, feature construction, and other learned transformations must be isolated from the evaluation data and, in an out-of-fold stacking procedure, must also be refitted within each training fold [5–7]. These considerations are especially important for small datasets, for which a modest number of observations can produce large changes in estimated performance.

The interpretation of tabular machine-learning models also requires caution. Explainable artificial-intelligence methods such as SHAP and LIME can describe which variables are influential for a fitted prediction function and how individual predictions are locally represented [21–23]. Accumulated Local Effects provide an additional way of visualising model behaviour while accounting for the empirical distribution of predictors [24]. However, attribution magnitude does not establish causality, and an explanation of a fitted model does not demonstrate that the underlying association is clinically meaningful. Recent reviews and reporting frameworks therefore emphasise explanation fidelity, stability, uncertainty, data provenance, calibration, external validation, and intended use when interpreting healthcare-oriented artificial intelligence systems [8, 9, 16, 17]. This distinction is particularly important for benchmark datasets whose outcome categories are not independently clinically adjudicated.

The revised study evaluates a two-learner Random Forest–XGBoost stack with a multinomial logistic meta-learner. Its methodological contribution is not a new classifier; it is a reproducible audit of a small public benchmark. The audit verifies literal outcome coding, detects duplicate predictor profiles, confines every learned transformation to the relevant training fold, generates meta-features out of fold, compares matched conventional baselines, and releases splits and observation-level probabilities. The target is explicitly the dataset's recorded sleep-status category, not a clinical diagnosis or a risk estimate.

Post-hoc explanation methods can describe the fitted function but cannot establish biological mechanisms. Because duplicate profiles, conflicting labels, and correlated predictors undermine stable attribution rankings here, the revised manuscript does not use SHAP, LIME, PCA, or ALE displays as substantive evidence. Instead, it reports the validation design, discrimination, probability scores, reliability curves, and full prediction artefacts required for audit.

2. Related work and study rationale

Table 1 summarises the related studies and their relevance to the present benchmark.

Recent sleep-AI research has concentrated primarily on apnea detection from physiological signals and wearable devices. Contemporary reviews cover electrocardiographic, oximetric, respiratory, and multimodal signal pipelines, and consistently distinguish automated screening from clinically adjudicated diagnosis [3, 4, 11, 12]. This evidence base is informative for clinical framing but is not directly comparable with the present low-dimensional lifestyle dataset, which contains neither raw physiological waveforms nor polysomnographic severity indices. For tabular prediction, recent evidence continues to treat tree ensembles and neural architectures as complementary rather than universally interchangeable model families [13]. Alom et al. applied ANN bagging, boosting, and weighted averaging to the same sleep-health benchmark used here [14]; their reported accuracies provide the principal literature-derived numerical comparator in this manuscript. In a different clinical setting, Alabi et al. demonstrated that heterogeneous stacking can be paired with SHAP and LIME, but their cancer-survival task, sample construction, and validation design preclude direct performance comparison [15].

Beyond single-model approaches, ensemble learning provides a practical means of combining models with different inductive biases on structured data. Bagging-based methods reduce variance through aggregation across decision trees, whereas boosting iteratively focuses model capacity on residual errors and can capture complementary nonlinear decision boundaries. Simple probability averaging combines model outputs but applies a fixed aggregation rule. Stacked generalisation instead learns a second-level mapping from base-model predictions, allowing the final predictor to adapt to complementary predictive patterns among the constituent learners [28]. For small structured datasets, however, the construction of the second-level training set is critical: using in-sample base-model predictions can expose the meta-learner to overly optimistic signals. Out-of-fold prediction generation provides a means of training the meta-learner on predictions produced for observations that were not used to fit the corresponding base learner. This motivates the duplicate-aware, out-of-fold RF–XGBoost stacking audit used here. The study compares stacking, soft voting, and conventional baselines on the same repeated grouped outer folds; its results are descriptive internal-validation estimates, not clinical performance claims.

3. Materials and Methods

3.1. Data audit and target definition

The analysis used the public Sleep Health and Lifestyle Dataset hosted on Kaggle [18]. The CSV was read with default missing-value conversion disabled. The target field was then stripped of surrounding whitespace and required to contain only the literal strings *None*, *Insomnia*, and *Sleep Apnea*; the program stops if an empty or undocumented outcome occurs. This audit found 219 literal *None* values, 77 *Insomnia* values, and 78 *Sleep Apnea* values, with no blank or missing target values. Thus, *None* was not created by imputing a blank field. Its clinical meaning is nevertheless unknown: it denotes a source-file category, not clinician-confirmed absence of disease.

Person ID was removed. "Normal Weight" was harmonised to "Normal", and Blood Pressure was parsed into systolic and diastolic numerical variables. After these deterministic operations, 265 rows duplicated a complete retained predictor profile, leaving 109 unique profiles. Twenty-two profiles had conflicting recorded outcome labels. Predictor-profile groups were therefore defined without using the outcome and were kept wholly within either development or evaluation data.

3.2. Models and duplicate-aware validation

Numerical predictors were standardised and categorical predictors one-hot encoded. Every fitted transformation was inside its model pipeline. The primary stack used Random Forest (150 trees; minimum leaf size 2), XGBoost (150 estimators; learning rate 0.03; depth 3; row and column subsampling 0.9), and a multinomial logistic meta-learner ($C = 1.0$). The meta-learner

Table 1
Summary of related studies and their relevance to this work.

Study	Data/task	Methods	Relevance
Alom et al. [14]	Sleep-health tabular data	ANN Ensembles	Main Numerical Comparator
Alabi et al. [15]	Cancer survival	Stacking, SHAP, LIME	Methodological context only
Bazoukis et al. [4]	Physiological sleep data	ML/DL methods	Clinical screening context
Tyagi et al. [11]	Sleep signals	ML/DL pipelines	Signal-based comparison
Borisov et al. [13]	Tabular datasets	Tree and neural models	Supports heterogeneous models
This study	**Sleep-Health Tabular Data**	**Heterogeneous stack**	**Grouped OOF stacking, calibration, uncertainty**

Table 2
Audit of the outcome field and retained predictor profiles.

Item	Count
Rows	374
Literal *None* outcomes	219
Literal *Insomnia* outcomes	77
Literal *Sleep Apnea* outcomes	78
Blank or missing outcomes	0
Unique retained predictor profiles	109
Rows duplicating a retained profile	265
Profiles with conflicting recorded outcomes	22

received only out-of-fold base-model class probabilities; encoded-feature pass-through was not used. These fixed settings were used without test-fold tuning.

Two shuffled repetitions (seeds 17 and 29) of five-fold StratifiedGroupKFold formed the outer evaluation. For every outer-training partition, four-fold StratifiedGroupKFold generated the base-model out-of-fold probabilities. The encoder, scaler, and each base learner were refitted within the corresponding inner training fold. Final base learners were then fitted on the entire outer training partition before outer-test prediction. The matched comparators were RF–XGBoost soft voting, Random Forest, XGBoost, logistic regression, a shallow decision tree, and histogram gradient boosting. All comparators used the same outer folds.

Figure 1 documents the executed analysis rather than the earlier 80:20 hold-out workflow. It makes explicit the duplicate-group boundary, the two repeated outer validations, four-fold inner out-of-fold meta-feature generation, absence of encoded-feature pass-through, and the prediction-level artefacts released by the script.

3.3. Reproducibility and evaluation

The executable analysis is provided as `revised_analysis.py`. It records software versions, seeds, group identifiers, outer-fold assignments, fold-level metrics, and observation-level reference labels, predictions, and class probabilities. Accuracy, macro-F1, multiclass log loss, and multiclass Brier score were calculated per outer fold. For stacking versus soft voting, we report the mean matched fold-level accuracy difference, a 10,000-resample percentile bootstrap interval, and an exact McNemar test on the first repetition's complete out-of-fold predictions. Repeated folds are correlated; these quantities describe internal-validation variability and do not represent external validation.

The earlier SHAP, LIME, PCA, and ALE graphics are not retained as evidence for the final stacking classifier. Instead, the executable analysis adds a deliberately limited, auditable TreeSHAP diagnostic for the XGBoost *base learner* only: five grouped outer-test folds provide feature-attribution stability estimates, and one held-out prediction includes its row index, observed label, predicted probabilities, and an additivity residual. This diagnostic does not explain the full stack, establish clinical mechanisms, or support deployment claims.

4. Results

Table 3 gives the ten outer-fold results. Logistic regression had the highest mean accuracy and macro-F1. Stacking had a slightly lower mean accuracy than soft voting, although its mean log loss and Brier score were modestly lower. Thus, the revised analysis does not support a claim that the stack outperforms either soft voting or logistic regression on discrimination.

The mean stacking-minus-soft-voting accuracy difference was -0.0031. Its bootstrap 95% interval was -0.0162 to 0.0070, which crosses zero. In the first repetition, stacking alone was correct for two observations and soft voting alone was correct for four; the exact McNemar p value was 0.6875. The small number of discordant observations limits power. These results establish neither a stacking advantage nor equivalence.

Figure 2 shows class-wise reliability curves from the first repetition's out-of-fold predictions. The curves and Brier scores are internal diagnostic summaries only; they do not demonstrate calibration in a new population.

4.1. Audited XAI diagnostics

To retain an explainability component without reintroducing unsupported claims, TreeSHAP was applied only to the fitted XGBoost base learner in the first repetition's five grouped outer-test folds. One-hot contributions were summed to their original predictor before ranking. Figure 3 reports mean absolute TreeSHAP values with the 10th–90th empirical percentile band across these five folds. Systolic blood pressure had the largest mean attribution, but its range was substantial; diastolic blood pressure changed from rank 1 to rank 9 across folds. These diagnostics therefore describe model dependence and its instability, not a stable biological ranking or a stacking-model explanation.

For the local audit, the script selected the highest-confidence XGBoost outer-test prediction in repetition 1, fold 1: CSV row 62 (zero-based), duplicate group 71, recorded label *None*, predicted label *None*, and predicted probability 0.959. Figure 4 gives its largest transformed-feature contributions. The associated raw-margin TreeSHAP additivity residual was 1.43×10^{-6}, recorded with all class probabilities in `xai_local_audit.json` and `xai_local_audit.csv`. Fidelity of this additive decomposition does not demonstrate causal validity, robustness to a different background distribution, or fidelity for the final stack.

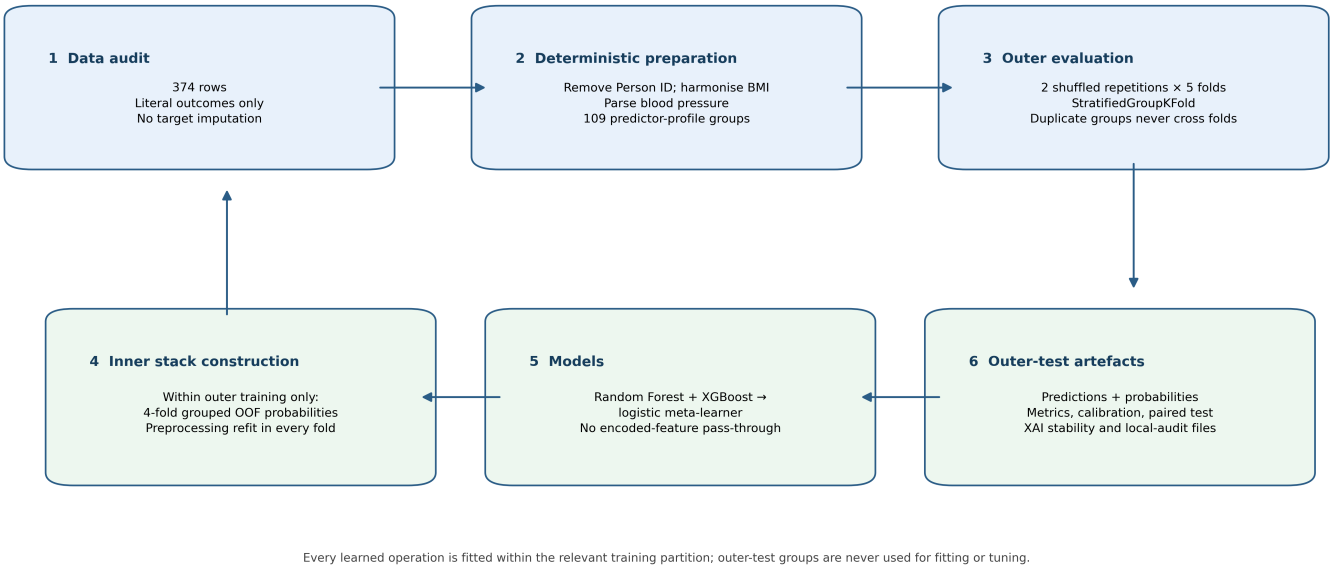


Figure 1. Executed duplicate-aware analysis architecture. Predictor-profile groups are created after deterministic preparation and remain intact in all outer and inner splits. Random Forest and XGBoost class probabilities are the only meta-learner inputs; encoded predictors are not passed through. Every learned transformation is fitted only within its relevant training partition.

Table 3
Duplicate-aware repeated outer-validation performance. Values are mean (standard deviation) across ten outer folds. Lower is better for log loss and Brier score.

Model	Accuracy	Macro-F1	Log loss	Multiclass Brier
Logistic regression	0.896 (0.032)	0.852 (0.073)	0.441 (0.117)	0.190 (0.048)
Shallow decision tree	0.858 (0.037)	0.807 (0.063)	0.933 (0.635)	0.227 (0.057)
Histogram gradient boosting	0.861 (0.067)	0.800 (0.083)	0.620 (0.211)	0.239 (0.094)
Random Forest	0.886 (0.034)	0.837 (0.060)	0.679 (0.331)	0.192 (0.049)
XGBoost	0.889 (0.032)	0.832 (0.070)	0.409 (0.108)	0.191 (0.057)
Soft voting (RF–XGBoost)	0.890 (0.033)	0.835 (0.078)	0.422 (0.105)	0.189 (0.051)
Stacking (RF–XGBoost)	0.887 (0.034)	0.832 (0.069)	0.397 (0.094)	0.187 (0.054)

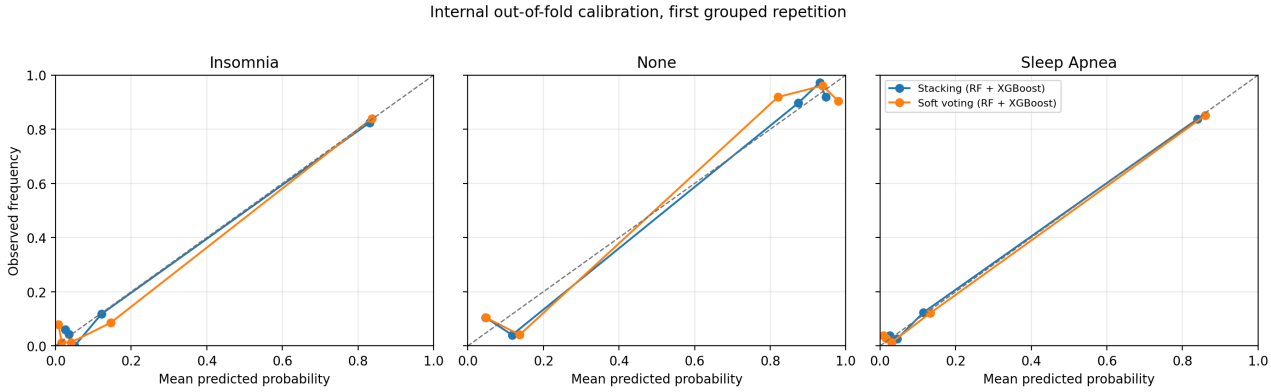


Figure 2. Internal class-wise reliability curves from the first duplicate-aware grouped-validation repetition. Curves are based on out-of-fold predictions and are descriptive; an external cohort is required to assess transportable calibration.

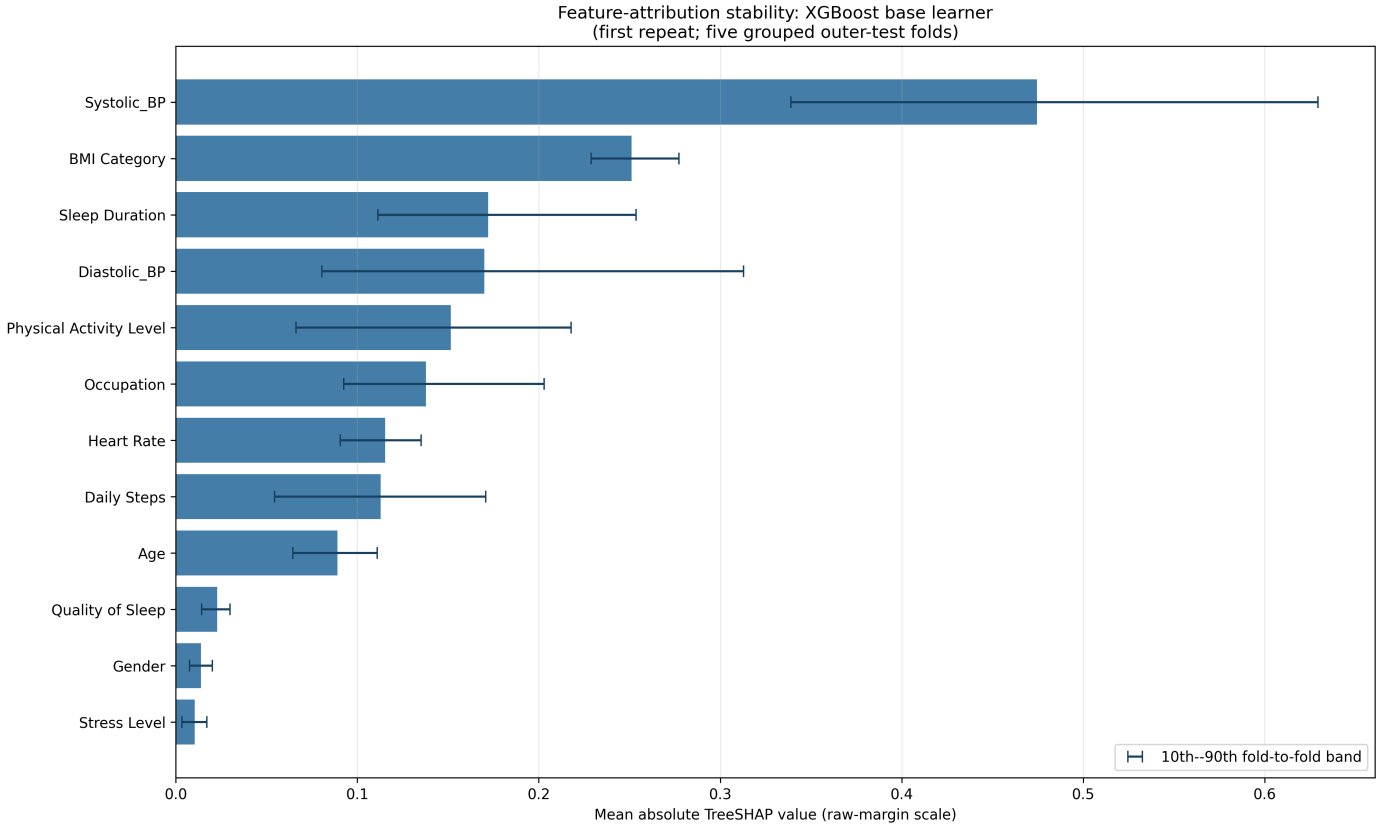


Figure 3. Feature-attribution stability for the XGBoost base learner, not the stacked classifier. Bars are mean absolute TreeSHAP values over five grouped outer-test folds from the first repetition. Error bars are the defined 10th–90th empirical percentile band; they show material fold-to-fold variability for several predictors.

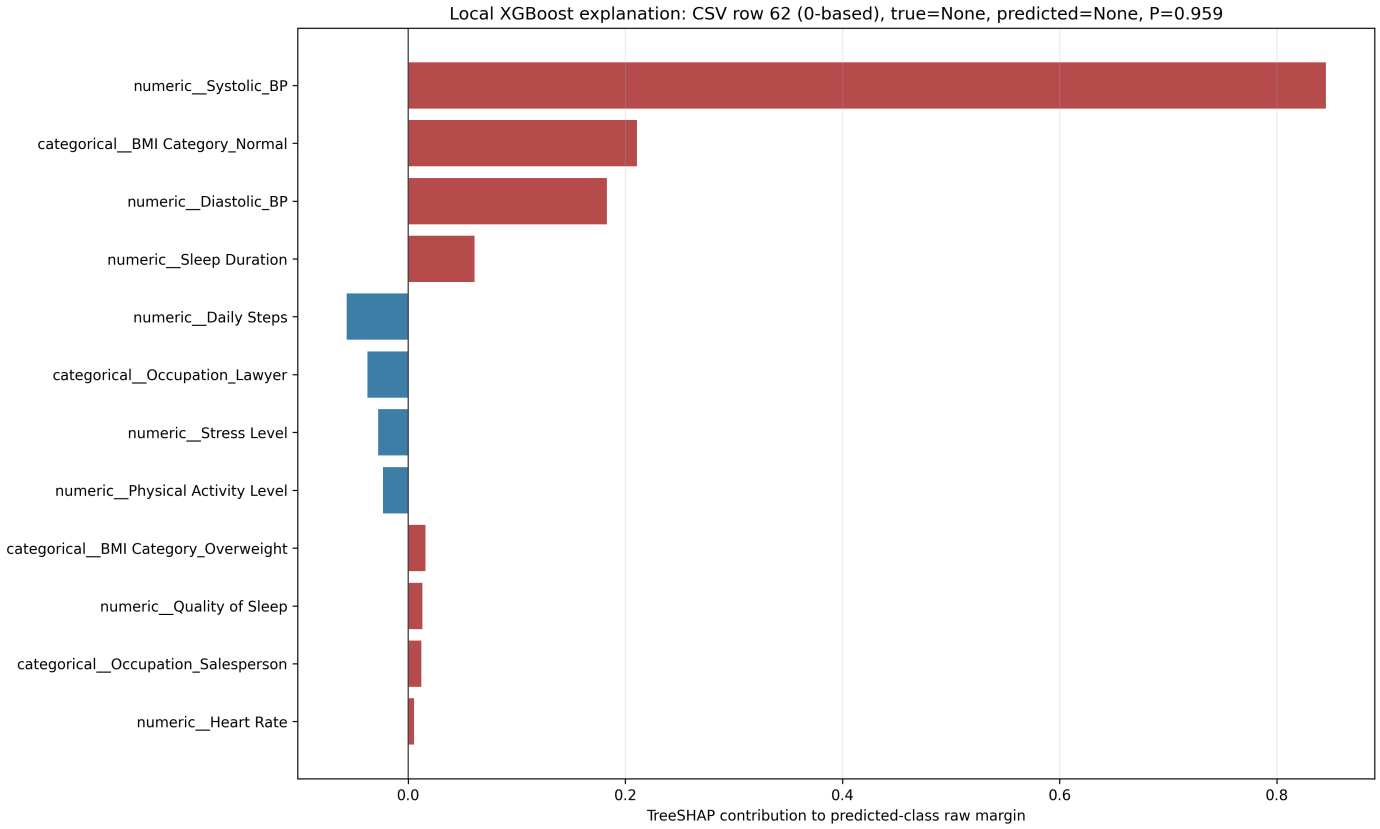


Figure 4. Audited local TreeSHAP diagnostic for an XGBoost base-learner outer-test prediction: CSV row 62 (zero-based), true label *None*, predicted label *None*, and predicted probability 0.959. Positive and negative bars change the predicted-class raw margin. The row identity, all three class probabilities, duplicate group, and additivity residual are released with the figure.

5. Discussion

The original single random hold-out estimate was sensitive to duplicate predictor profiles crossing the evaluation boundary. Under duplicate-aware validation, the observed performance is lower and more variable, and logistic regression is competitive with the ensemble. This does not invalidate the computational exercise; it changes its interpretation. The principal contribution is an auditable evaluation protocol for a small, duplicate-rich benchmark, not a new stacking algorithm or evidence of superiority over established models.

The recorded outcome is not a clinically adjudicated diagnostic standard. Sleep duration and quality may also overlap with information used to identify insomnia, and 22 identical predictor profiles have discordant recorded labels. Accordingly, performance on this dataset must be separated from the requirements of real-world clinical screening: a defined intended use, independently adjudicated reference outcomes, prospective evaluation, subgroup assessment, calibration in the target population, and external validation. The XGBoost attribution figures describe one fitted base learner, not a biological mechanism or the decision logic of the complete stack; feature dependence can redistribute importance among correlated predictors and across ensemble architectures.

6. Limitations

The study uses one small public dataset with undocumented collection provenance and no external clinical cohort. The literal *None* category may still introduce classification bias because its clinical semantics are not established. Repeated grouped folds share the same source dataset and are not independent replications. Model settings were fixed rather than optimised through a broad hyperparameter search. The reported probability and calibration summaries are internal only. The XAI diagnostic concerns the XGBoost base learner rather than the stack, and five correlated folds provide only an internal stability screen. These limitations preclude diagnostic, treatment, or deployment claims.

7. Conclusion

This revision retains RF–XGBoost stacking as the primary computational configuration but evaluates it with verified outcome coding, duplicate-aware fold assignment, fold-specific preprocessing, matched comparators, released prediction-level artefacts, and a bounded XGBoost base-learner XAI audit. The stack did not exceed soft voting or logistic regression in mean accuracy or macro-F1. The work should therefore be interpreted as a reproducible benchmark audit of recorded sleep-status labels, not as validated clinical risk stratification or diagnostic evidence.

Author Contributions

Rifat Bin Reza: Conceptualization, Methodology, Data curation, Software, Formal analysis, Investigation, Model development, Experimental design, Visualization, Writing–original draft, Writing–review and editing, Project administration. Md Ahadul Haque Suvo: Software, Methodology, Data preprocessing, Model implementation, Validation, Formal analysis, Investigation, Visualization, Writing–review and editing. Atair Rahman Alvi: Investigation, Validation, Writing–review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

This study involved secondary computational analysis of the publicly available Sleep Health and Lifestyle Dataset on Kaggle [18]. No participants were recruited or contacted, no interventions were performed, and no new human or animal data were collected. The administrative Person ID field was excluded before modelling.

Data availability

The Sleep Health and Lifestyle Dataset is publicly available on Kaggle at https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset. Dataset reuse is subject to the source platform's licence and terms.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude 3.5 Sonnet to assist with manuscript readability, grammar and spelling, literature-search planning, and LaTeX formatting. After using the tool, the authors reviewed and edited the output as needed and take full responsibility for the accuracy, integrity, and final content of the manuscript.

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